

CMPG 325 - Computer Networks

Milestone 1

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Date of Submission	28 August 2026

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CLIENT DESIGN REVIEW

A. CLIENT REQUIREMENTS

I. BACKGROUND

The assigned client for this project is Leano Gaming Lounge, an entertainment establishment located in Mahikeng. The organisation operates as a venue where customers frequent to utilise gaming consoles and entertainment services. The business also conducts administrative operations, maintains security surveillance systems, and has identified the need to provide guest Wi-Fi services to enhance the customer experience.

II. BUSINESS REQUIREMENTS

- Provide reliable connectivity for gaming stations, office computers, security cameras and a server.
- Allow staff and the manager to share a server and reach the Internet.
- Keep security camera (CCTV) traffic separate from office data traffic.
- Offer visitors Wi-Fi without exposing any internal resource.
- Use the assigned address block 192.168.55.0/24 efficiently, with room for growth.

FUNCTIONAL REQUIREMENTS

- **FR1** - Five gaming PCs must obtain addresses automatically and reach the Internet.
- **FR2** - 3 office PCs (2 staff and 1 manager) must obtain addresses automatically, reach the server and reach the Internet.
- **FR3** - Five cameras (two inside, three outside) must have fixed addresses and be reachable only from permitted hosts.
- **FR4** - One server must provide DHCP and DNS to the internal networks.
- **FR5** - One router must provide inter-VLAN routing and traffic filtering.
- **FR6** - Two switches must be connected so that VLANs span both switches.
- **FR7** - One access point must provide guest Wi-Fi.
- **FR8** - Visitors must reach the Internet but no internal subnet.

III. DESIGN CONSTRAINT

CCTV traffic must be segmented from office data traffic. CCTV devices are placed in their own VLAN and subnet, and the router blocks traffic between the CCTV network and the Office and Gaming networks in both directions. The only internal destination CCTV may reach is the Servers subnet, which represents recording or monitoring.

IV. ASSIGNED NETWORKING CHALLENGE: VLAN TRUNKING (802.1Q)

VLAN trunking must be configured, verified and explained across the two switches. The design makes sure that Gaming, Office and CCTV VLANs each have members on both switches, so the trunk carries real traffic and can be proven by same-VLAN pings across the trunk.

V. CHANGE REQUEST CR3: GUEST WI-FI

Guest Wi-Fi is added through the access point on its own VLAN (VLAN 40) and subnet. Guests receive addresses from the router, use an Internet-side DNS server, and are blocked by an access list from every internal subnet. The scope stays exactly as worded in the change request, with no further additions.

B. TOPOLOGIES AND IP ADDRESSING

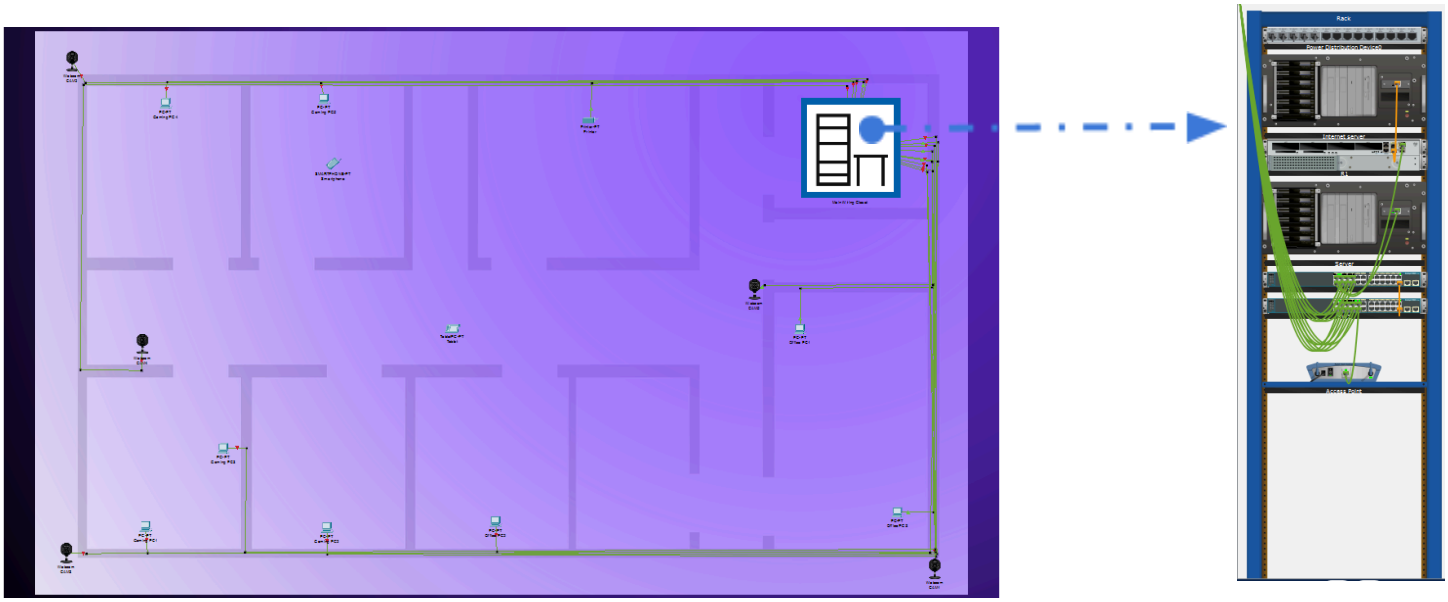
I. DEVICE INVENTORY

Device	Qty	Packet Tracer model	Purpose
Office PCs	3	PC1 to PC3	2 staff and 1 manager
Printer	1	Printer	
Gaming PCs	5	PC1 to PC5	Customer gaming stations
Cameras	5	CAM1 to CAM5	2 indoor, 3 outdoor
Access point	1	AccessPoint-PT	Guest Wi-Fi
Server	1	Server-PT	DHCP and DNS for internal VLANs
Router	1	Cisco 2911	Inter-VLAN routing, ACLs, guest DHCP
Switches	2	Cisco 2960 (SW1, SW2)	Access layer, joined by 802.1Q trunk
Guest device	2	Smartphone and tablet	Wireless test devices for CR3
Internet server	1	Server-PT	Simulated Internet and external DNS

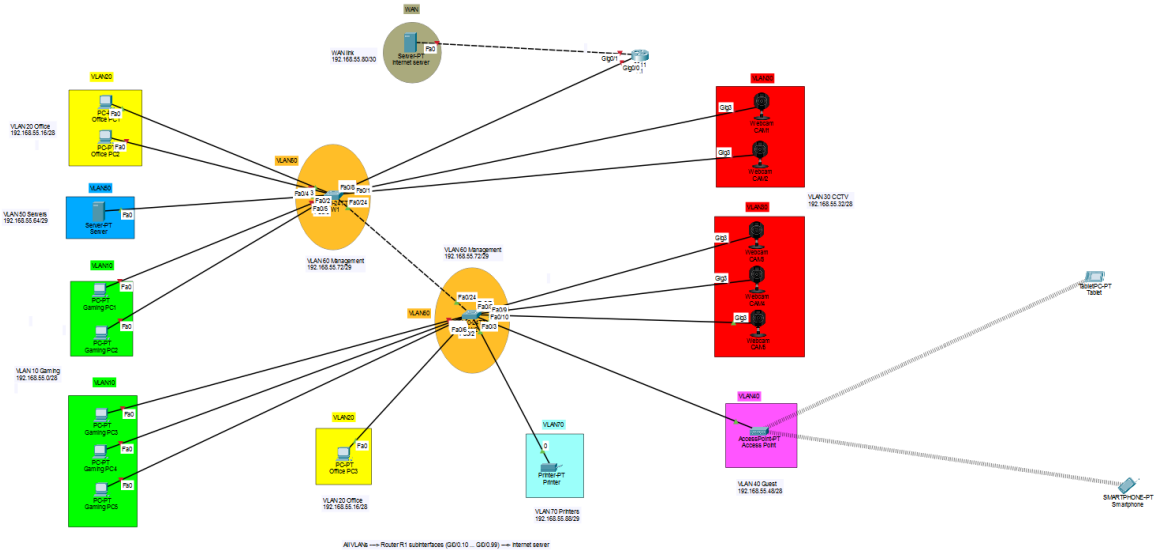
II. CABLING AND PORT PLAN

From	Device Port	To	Device Port	Cable
R1	Gi0/0	SW1	Fa0/1	Copper straight-through
R1	Gi0/1	Internet server	Fa0	Copper cross-over
SW1	Fa0/24	SW2	Fa0/24	Copper cross-over
SW1	Fa0/2	Server	Fa0	Copper straight-through
SW1	Fa0/3 to Fa0/4	Office PC1, PC2	Fa0	Copper straight-through
SW1	Fa0/5 to Fa0/6	Gaming PC1, PC2	Fa0	Copper straight-through
SW1	Fa0/7 to Fa0/8	CAM1, CAM2	Gig3	Copper straight-through
SW2	Fa0/2 to Fa0/3	Office PC3, PC4	Fa0	Copper straight-through
SW2	Fa0/4 to Fa0/6	Gaming PC3, PC4, PC5	Fa0	Copper straight-through
SW2	Fa0/7 to Fa0/9	CAM3, CAM4, CAM5	Gig3	Copper straight-through
SW2	Fa0/10	Access point	Port 0	Copper straight-through

III. PHYSICAL TOPOLOGY



IV. LOGICAL TOPOLOGY



V. IP ADDRESSING TABLE

VLAN	Interface	Network	Mask	Usable range	Hosts needed	Capacity	Address assignment
10	Gaming	192.168.55.0	/28 (255.255.255.240)	.1 to .14	5	14	DHCP from server, pool .2 to .14
20	Office	192.168.55.16	/28 (255.255.255.240)	.17 to .30	4	14	DHCP from server, pool .18 to .30
30	CCTV	192.168.55.32	/28 (255.255.255.240)	.33 to .46	5	14	Static, .34 to .38
40	Guest	192.168.55.48	/28 (255.255.255.240)	.49 to .62	Up to 13	14	DHCP from router, pool .50 to .62
50	Servers	192.168.55.64	/29 (255.255.255.248)	.65 to .70	1	6	Static, server .66
60	Management	192.168.55.72	/29 (255.255.255.248)	.73 to .78	2	6	Static, SW1 .74, SW2 .75
70	Printer	192.168.55.88	/29 (255.255.255.248)	.89 to .94	1	6	Static, printer .90
99	Native	none	none	none	0	none	Unused
WAN	Router to Internet	192.168.55.80	/30 (255.255.255.252)	.81 to .82	2	2	Static, router .81, Internet server .82

C. SECURITY DESIGN

I. CCTV SEGMENTATION (DESIGN CONSTRAINT)

The filters are placed on the router subinterfaces of CCTV, Office and Gaming, because access lists are stateless and so must be applied on each side to block traffic cleanly.

Source	Destination	Result
CCTV VLAN 30	Servers VLAN 50	Permitted (recording or monitoring)
CCTV VLAN 30	Office VLAN 20 and Gaming VLAN 10	Blocked
Office VLAN 20	CCTV VLAN 30	Blocked
Gaming VLAN 10	CCTV VLAN 30	Blocked

II. GUEST ISOLATION (CR3)

Source	Destination	Results
Guest VLAN 40	Any internal subnet (192.168.55.0 to .79)	Blocked
Guest VLAN 40	Internet server (192.168.55.82)	Permitted
Guest VLAN 40	Router DHCP (UDP 67 and 68)	Permitted

- Unused switch ports shut down and placed in a parking VLAN.
- A dedicated management VLAN (60) for switch administration.
- Native VLAN moved off VLAN 1.
- Passwords and a login banner on the router and switches.

III. DESIGN DECISIONS AND JUSTIFICATION

Decision	Justification
VLANs instead of one flat network	Separates gaming, office, CCTV and guests for security and smaller broadcast domains.

Router-on-a-stick	Low cost and enough for a small lounge with one router.
Two switches with spread VLAN members	Lets the trunk be demonstrated with real cross-switch traffic.
VLSM addressing	Fits all subnets into the 192.168.55.0/24 block, with room left to grow.
/29 for Servers and Management	Only a few devices are needed in each, so smaller subnets avoid waste.
Router-provided DHCP for guests	Guests need no access to the internal server, which keeps CR3 isolation clean.
Static addresses for cameras	Cameras need predictable addresses for recording and access control.
Native VLAN 99 and allowed VLAN list	Reduces VLAN hopping and limits what the trunk carries.
Extended ACLs on both sides of the CCTV boundary	ACLs are stateless, so blocking one direction only would give inconsistent results.
PCs used to simulate cameras	Simpler and more reliable in Packet Tracer than IoT registration, with identical traffic behaviour for testing.

IV. TESTING PLAN

Test	**Expected result**
Gaming PC1 (SW1) to Gaming PC3 (SW2)	Success, same VLAN across the trunk
Office PC1 (SW1) to Office PC3 (SW2)	Success, same VLAN across the trunk

Server to CAM4 (on SW2)	Success, reachable through the router and the trunk
Office PC to Server	Success, inter-VLAN
Office or Gaming PC to any camera	Fail by design
Camera traffic to Office or Gaming (seen in CCTV-IN deny counters)	Fail by design
Guest device to any internal subnet	Fail by design
Gaming PC or Guest device to Printer	Fail by design
Guest device to Internet server	Success
Office and Gaming PCs to Internet server	Success
DHCP lease on every DHCP VLAN	Correct address, gateway and DNS
DNS lookup from an office PC	Resolves correctly

D. SOFTWARE TOOL AND PROJECT PROFILE

I. **Cisco Packet Tracer:** The primary simulation environment used to build the physical and logical topology, configure the 802.1Q trunking protocols, and perform end-to-end connectivity testing.

II. **Repository Link:** https://github.com/MateyeSelota/CMPG325-2026-125